

TUTORIAL3

3) Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

INPUT:

```
#include <iostream>
```

```
using namespace std;
```

```
void swap(int &a, int &b) {  
    int temp = a;  
    a = b;  
    b = temp;  
}
```

```
void readArray(int arr[], int size) {  
    cout << "Enter " << size << " numbers: ";  
    for (int i = 0; i < 4; i++) {  
        cin >> arr[i];  
    }  
}
```

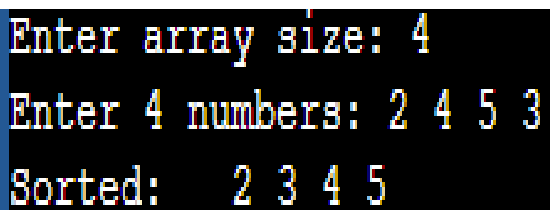
```
void displayArray(int arr[], int size) {  
    for (int i = 0; i < 4; i++) {  
        cout << arr[i] << " ";  
    }  
    cout << endl;  
}
```

```
void sortArray(int arr[], int size) {  
    for (int i = 0; i < 4 - 1; i++) {  
        for (int j = 0; j < 4 - i - 1; j++) {  
            if (arr[j] > arr[j + 1]) {  
                swap(arr[j], arr[j + 1]);  
            }  
        }  
    }  
}
```

```
int main() {  
    int size;
```

```
cout << "Enter array size: ";  
cin >> size;  
  
int arr[4];  
  
readArray(arr, 4);  
  
sortArray(arr, 4);  
  
cout << "Sorted:  ";  
displayArray(arr, 4);  
  
return 0;  
}
```

OUTPUT:

A screenshot of a terminal window with a black background and yellow text. The output shows three lines: 'Enter array size: 4', 'Enter 4 numbers: 2 4 5 3', and 'Sorted: 2 3 4 5'.

```
Enter array size: 4  
Enter 4 numbers: 2 4 5 3  
Sorted: 2 3 4 5
```